I've included an example in `counting-tokens.google.mjs` that demonstrates this approach. To run it, follow these steps:

1. Install the dependencies:

```
npm install
```

2. Set environment variables, and run the script:

```
export GOOGLE_CLOUD_PROJECT="your-project-id"
export GOOGLE_CLOUD_LOCATION="your-location"
node counting-tokens.google.mjs
```

CONTROL TOKENS

Control tokens (or special tokens) are basically ways to control and guide the behavior, format, or content of the LLM's output from within prompts. In simple terms, we use special marks or keywords to define prompt regions that the LLM needs to apply certain rules and instructions.

There are various types of control tokens depending on the LLM we use. A few example types from the OpenAI GPT-4 model are

- *End-of-text token* (`<|endoftext|>` *or* `⟨|endoftext|⟩`)—Marks the end of a sequence of text
- *Beginning-of-sequence token* (`<|startoftext|>` *or* `⟨|startoftext|⟩`)—Marks the beginning of a sequence of text
- `<|padding|>`—Used for padding sequences to a fixed length when necessary

The exact tokens can vary depending on the model architecture, the version, and the specific implementation or fine-tuning of the model. For example, the following is the raw user prompt that includes various special tokens commonly associated with OpenAI's GPT-4 in a chat-like environment:

```
<|startoftext|>
<|user|> What is the capital of France?
<|assistant|> The capital of France is Paris.
<|endoftext|>
```

> **NOTE** Control tokens are handled internally by the API. When interacting with the OpenAI API, you don't need to worry about explicitly using special tokens like `<|startoftext|>`, `<|endoftext|>`, or role indicators such as `<|user|>` and `<|assistant|>`. These tokens are managed internally by the model to organize and process the conversation, ensuring seamless interaction. The API abstracts away these details, allowing you to send and receive plain-text messages without needing to manually handle these tokens. While you don't need to manually use control tokens when working with the OpenAI API, it's helpful to understand their purpose—especially when debugging, inspecting raw prompt logs, or fine-tuning models where you may encounter them explicitly.